

Education

B.S. Mechanical Engineering, Tau Beta Pi, University of California, Berkeley

Exp Graduation: May 2028 | GPA: 3.98

Relevant Experience

Manufacturing & Automation Engineering Intern (Incoming) • 10x Genomics

Summer 2026

- Accepted offer to join Manufacturing & Automation Engineering team to support design, commissioning, and continuous improvement of production equipment and automated test systems.

Manufacturing Lead, Chassis Engineer • Formula Electric at Berkeley

Sept 2024 - Present

- Led design and manufacturing of 2025 steel spaceframe, running 30+ ANSYS FEA iterations to optimize tube geometry and node placement, improving stiffness-to-weight ratio by 25%
- Designed and built torsional stiffness test rig, validating FEA within <15% error against physical measurements
- Built Python sensitivity models linking chassis torsional stiffness to suspension compliance and LLTD, guiding design decisions on handling and tunability
- Executed explicit dynamic simulations in LS-DYNA to validate composite anti-intrusion plates for FSAE crashworthiness compliance; modeled non-linear failure modes to ensure rule compliance under load
- Conducted ASTM-standard 3-point bend and dye penetrant tests for rule-compliant welded and composite samples

Lead Research Engineer • Theoretical & Applied Fluid Dynamics Laboratory (TAFLAB)

May 2025 - Present

- Conducted multiphysics optimization in ANSYS Maxwell for a low-frequency (1Hz) electromagnetic harvester; achieved 200mW avg power, satisfying 20% of the total power requirement for autonomous surface vehicle (ASV) sensor suites.
- Prototyped and optimized oscillating electromagnetic energy harvester in ANSYS Maxwell to <6.3% error
- Modeled oscillatory system dynamics in MATLAB with Lagrangian mechanics to select spring stiffness for resonance tuning
- Created custom coil winding jig and rectification circuitry to convert AC output into regulated DC for battery charging

Propulsion Manufacturing Engineer • UC Berkeley's Space Technology and Rocketry (STAR)

Sept 2024 - May 2025

- L2 Rocket: Designed and fabricated with integrated avionics, achieving successful launch and recovery at 2,944 ft
- Liquid Engine: Applied DFM to manufacture 9 custom injector and combustion chamber components; used GD&T to ensure strict precision and tolerances for CNC/mill/lathe for successful static hot fire.

VP of Mechanical Design • Fremont High Robotics FRC Team 3501: Firebots

2020 - 2024

- Led a 6-person CAD team using Onshape, iterating a competition robot that placed in top 15% of 3,000+ teams worldwide.
- Managed all parts procurement (BOM), technical drawing approval, training program, and manufacturing workflow.

Personal Projects

ENGIN 100 - Electronics for the Internet of Things - Automatic Rice Cooker

- Developed an IoT rice cooking system using Onshape and Creaform 3D scanning to design custom mechanical interfaces
- Integrated ESP32 microcontrollers with ultrasonic and load cell sensors to automate rice portioning, washing, and cooking

ENGIN 29 - Manufacturing and Design Communication - Bike Trailer (Best Design & Integration Award)

- Designed and manufactured a collapsible bike trailer, reducing to ¼ of its original footprint for storage.

Combat Robotics

- Designed, built, and tested multiple fully 3D-printed 1 lb combat robots for local competitions, modeling an offset slotted-link shuffler drivetrain mechanism in MATLAB and prototyping the crank-and-slider-based design.

Skills

CAE/Analysis: ANSYS, LS-DYNA, MATLAB, Python | **Design:** Onshape, SolidWorks, GD&T, DFM | **Shop:** TIG, CNC, Mill, Lathe